

Repack & Sign APK – Complete Guide

APK modify karne ke baad sab se important step hai use wapas repack (rebuild) karna aur sign karna. Bina sign kiye APK install nahi hoti kyunke Android har APK ko digital signature require karta hai. Yeh guide aapko step-by-step repack aur sign karne ka complete process batayegi.

Repack Kya Hai?

Repack ka matlab hai decompiled APK ko wapas APK format mein convert karna. APKTool b (build) command se yeh kaam karti hai. Yeh modified smali code, resources, aur AndroidManifest.xml ko wapas DEX files aur compiled resources mein convert karti hai.

Sign Kya Hai?

Sign ka matlab hai APK ko digital certificate ke saath sign karna. Android har APK ko sign hone ki zaroorat hoti hai taake:

- App ki integrity verify ho (modify nahi hui)
- App ka developer identity establish ho
- APK install ho sake

Step 1: APK Rebuild (Repack) Karna

Pehle APKTool se decompiled folder ko wapas APK mein build karein:

```
apktool b decompiled/ -o modified.apk
```

Yahan:

- decompiled/ – Decompiled folder ka naam (jo apktool d se bana tha)
- modified.apk – Output APK ka naam

Example:

```
apktool b app-decompiled/ -o app-modified.apk
```

Build successful ho toh output folder mein modified.apk file create ho jati hai.

Common Issues:

- Agar resources mein koi error hai toh build fail ho sakti hai
- Smali syntax error bhi build fail kar sakti hai
- Error messages ko read karein aur fix karein

Step 2: Debug Signature Generate Karna

Sign karne se pehle aapko ek keystore file create karni hogi. Yeh ek digital certificate hai jo aapki identity represent karega.

Debug keystore (testing ke liye) generate karein:

```
keytool -genkey -v -keystore debug.keystore -alias  
debug -keyalg RSA -keysize 2048 -validity 10000
```

Yahan:

- -genkey – Key generate karein
- -v – Verbose output
- -keystore debug.keystore – Keystore file ka naam
- -alias debug – Alias (identifier)
- -keyalg RSA – Algorithm
- -keysize 2048 – Key size
- -validity 10000 – Certificate validity (dinon mein)

Questions:

- Password enter karein (yaad rakhein, baad mein zaroorat hogi)
- First and Last Name: Test user type karein
- Organizational Unit: Test type karein
- Organization: Test type karein
- City: Test type karein
- State: Test type karein
- Country Code: US type karein

Step 3: APK Sign Karna

Ab jarsigner tool use karke APK sign karein:

```
jarsigner -verbose -sigalg SHA1withRSA -digestalg  
SHA1 -keystore debug.keystore app-modified.apk  
debug
```

Yahan:

- -verbose – Detailed output
- -sigalg SHA1withRSA – Signature algorithm
- -digestalg SHA1 – Digest algorithm
- -keystore debug.keystore – Keystore file
- app-modified.apk – APK file
- debug – Alias (jo pehle set kiya tha)

Password: Keystore ka password enter karein (jo pehle set kiya tha).

Alternative (Android SDK ke zariye):

```
apksigner sign --ks debug.keystore app-  
modified.apk
```

Yeh automatically SHA-256 use karta hai aur zyada secure hai.

Step 4: APK Verify Karna

Sign successful hui ya nahi check karne ke liye:

```
jarsigner -verify -verbose -certs app-modified.apk
```

Yeh command APK ki signature details display karegi. Agar "jar verified" message aata hai toh sign successful hai.

Android SDK ke zariye verify:

```
apksigner verify app-modified.apk
```

Step 5: APK Install Karna

Sign hone ke baad APK install kar sakte hain:

```
adb install app-modified.apk
```

Agar

"INSTALL_PARSE_FAILED_INCONSISTENT_CERTIFICATES" error aata hai toh pehle original app uninstall karein, phir install karein.

Alternative (Direct install):

```
adb install -r app-modified.apk
```

-r flag se reinstall (replace) hota hai.

Production Signing (For Release)

Agar aap app publish karna chahte hain toh production keystore use karein:

Step 1: Production Keystore Generate Karein

```
keytool -genkey -v -keystore my-release-key.keystore -alias my-alias -keyalg RSA -keysize 2048 -validity 10000
```

Step 2: APK Sign Karein

```
jarsigner -verbose -sigalg SHA1withRSA -digestalg SHA1 -keystore my-release-key.keystore my-app.apk my-alias
```

Step 3: Align APK (Optimization)

```
zipalign -v 4 my-app.apk my-app-aligned.apk
```

Zipalign APK ko optimize karti hai taake installation faster ho aur memory usage kam ho.

Complete Workflow Summary

1. Decompile:

```
apktool d app.apk -o decompiled/
```

2. Modify smali/resources: Text editor mein changes karein.

3. Rebuild:

```
apktool b decompiled/ -o modified.apk
```

4. Generate Keystore (agar pehle se nahi hai):

```
keytool -genkey -v -keystore debug.keystore -alias debug -keyalg RSA -keysize 2048 -validity 10000
```

5. Sign APK:

```
jarsigner -verbose -sigalg SHA1withRSA -digestalg SHA1 -keystore debug.keystore modified.apk debug
```

6. Verify:

```
jarsigner -verify -verbose -certs modified.apk
```

7. Install:

```
adb install modified.apk
```

Common Errors & Solutions

"Apktool: brut.common.BrutException: could not exec"

- Resources mein koi error hai, XML syntax check karein

"jarsigner: Certificate chain not found"

- Alias galat hai, correct alias use karein

"INSTALL_PARSE_FAILED_NO_CERTIFICATES"

- APK sign nahi hui, dobara sign karein

"INSTALL_PARSE_FAILED_INCONSISTENT_CERTIFICATES"

- Pehle original app uninstall karein

"INSTALL_FAILED_UPDATE_INCOMPATIBLE"

- Signature mismatch, pehle original app uninstall karein

 **Educational Purpose Only:** Yeh guide sirf educational aur learning purposes ke liye hai. Repack aur sign APK ka istemal sirf apne own apps aur authorized environments mein karein. Kisi doosre ke app ko bina permission ke modify, repack, ya redistribute karna applicable cyber laws ke under punishable hai.

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SYED GHOUS
